# Documentation for Audio and Image Augmentation Tool

This project is an image/audio augmentation tool for automating QA test-case generation, built as a rebuild of an earlier site that kept crashing under load, as part of a research extension under Prof. Jerry Gao.

It's split into two paths that do different jobs: the **local pipeline** ( `run.py` ) handles full dataset-scale batch augmentation, including unzipping an entire dataset, apply every enabled transform from `image_config.yaml` / `audio_config.yaml` , and writing hundreds of test cases to disk (work that needs CPU/RAM a free host can't guarantee). The **hosted Streamlit app** ( `app.py` ), by contrast, is a lightweight, preview-only experience: upload a single image or clip and see every transform applied to it in-memory so each session's compute stays small and predictable even with many classroom users hitting it at once on Streamlit Community Cloud's free tier.

---

## Run locally (for the batch augmentation)

1. pip install -r requirements.txt
2. Drop your dataset zip into `uploads/` ,
   named `image_dataset.zip` and/or `audio_dataset.zip`
3. `python run.py`
4. Find results in `data/augmented/image/` and `data/augmented/audio/`

# Run the hosted preview (Streamlit)

1. Visit Augmentation Test Cases · Streamlit

---

## Definitions

### Audio Augmentations

Level Augmentations
Transformations that modify the representation level of the audio signal, either in the time domain of the waveform (raw audio) or frequency domain (spectrogram)

- Gain Shift: randomly boosts/cuts overall volume
  - The applied gain is randomly chosen from +- 6dB around the original level
- Normalization: peak-normalizes audio so the loudest sample point hits 0dBFS (full scale)

- Clipping: simulates clipped/distorted audio by hard-limiting the waveform
    - The clipping threshold is picked based on the amplitude distribution of the signal and the threshold's percentile (for example, how much of the signal is left untouched) is randomly chosen between the minimum and maximum percentile. In short, low percentile values mean more aggressive clipping and chopping of peak amplitudes.

Noise Augmentations
Noise is how fast the energy rolls off as frequency rises.

- White Noise: adds flat-spectrum noise directly at an amplitude level
    - The noise's amplitude (it's relative to a normalized [1, 1] signal) is randomly sampled in the range between the minimum and maximum amplitude. It typically sounds like TV static, or a bright hiss.
- Pink Noise: noise geared towards lower frequencies, added relative to signal strength
    - Pink Noise controls the loudness of noise relative to the signal via Signal to Noise Ratio (SNR). A lower SNR indicates louder and more intrusive noise, while higher SNR (e.g. 40 dB) is barely audible. The SNR used per sample is randomly drawn from the selected range. Typically sounds like a steady rhythm, like rain or a crowd murmur. Consider Pink Noise the "middle" of White and Brown noise.
- Brown Noise is similar to Pink Noise, and typically sounds deep or muffled.

Temporal Augmentations

- Time Stretch: changes the playback speed without changing the pitch
    - It randomly stretches or compresses duration by a factor within the minimum and maximum rate selected (0.9 would be slower and 10% longer, 1.1 would be faster and 10% shorter)
- Time Shift
    - Rotates the audio (think of a circular conveyor belt) a random amount up to the max selected percent of the clip's total length in either direction.

Spectral Augmentations

- Pitch Shift: shifts the pitch randomly by up to +- the maximum semitone
- EQ Filtering: applies a multi-band equalizer
    - The audio is split into 7 frequency bands, and each band independently gets a random gain adjustment within +- the selected dB, simulating varied frequency response
- Bandstop Filter: a band-reject filter that removes a chunk of frequencies (simulating for example, phonelines)
    - The center frequency of the rejected band is randomly chosen between the selected minimum and maximum Hz

Codec

- MP3 Compress: simulates lossy compression artifacts by encoding/decoding through MP3
    - randomly picks a bitrate in a selected range, simulating varying degrees of audio compression and quality.

---

**Image Augmentations**

Transformation Augmentations

- Hue Shift: hue shifted randomly within +- selected degree (out of 360)
    - Commonly used to simulate different lighting color casts or making a model invariant to object color (e.g. red v.s. blue cars)
- Saturation Shift: adjusts color intensity/vividness
    - Saturation scaled by a random factor within +- selected % of original. Scaled from [0, 1], and is frequently used to improve robustness to washed out or oversaturated camera sensors.
- Brightness Shift: adjusts overall brightness
    - Brightness is shifted up by +- selected % scaled [0, 1] and is frequently used to simulate different types of day or exposure differences
- Contrast Shift: adjusts difference between light/dark areas
    - Contrast is shifted up by +- selected % scaled [0,1] Is used to improve robustness against hazy or sharp image values
- Gamma Shift: applies a nonlinear brightness curve
    - Applies a gamma value randomly picked between the gamma limit selection, where values below 100 brightens midtones and values above 100 darkens them. It's frequently used to mimic different display/sensor gamma curves
- RGB Shift: shifts each color channel independently
    - Each channel is offset by up to +- the selected %, and is independently randomized. It's frequently used to simulate color calibration differences
- Clahe: boosts local contrast rather than global, and is frequently used to fix uneven lighting. Can be used to recover detail in shadows/highlights
    - Caps how much contrast enhancement is allowed per local region before clipping within the selected limit
- White Balance Shift: simulates incorrect camera color-temperature calibration
    - Shifts the warm/cool balance by up to +- the selected %. Used to improve robustness to indoor lighting versus daylight versus shadow color casts

Obscure Augmentations:

- Gaussian Blur: smooths the image with a gaussian kernel
  - Kernel size is randomly chosen by a selected blur limit and is used to simulate out of focus or low quality images
- Motion Blur: directional blur simulating movement
  - Kernel size is also randomly chosen up to a selected blur limit and is applied along a random direction. Is used to simulate fast moving subjects
- Gaussian Noise: adds random pixel intensity noise
  - Randomly samples noise variance between the bounded limits and is used to simulate low light grain
- JPEG Compress: re-encodes through lossy JPEG compression
  - Randomly samples compression quality between the bounded limits and is used to improve robustness to compressed images from the web, media, etc
- Cutout: masks out random rectangular regions
  - Cuts out the selected number of holes and each are sized around the selected size's % of the relevant dimension. It forces models to not over rely on a single region
- Resize Degrade: downscales then upscales, losing resolution/detail
  - Randomly down-samples to as little as the selected % of the original size before restoring to original dimensions. It's frequently used to simulate low resolution source images

---

Geometry Augmentations

- Rotate: rotates the image
  - The rotation angle is randomly chosen within +- the selected limit and is used for orientation invariance or camera tilt simulation
- Crop: crops to a random sub-region
  - The crop area randomly covers between the bounded % limits of original image area. It's frequently used to simulate different zoom levels or object positions
- Zoom: scales image in and out
  - Zoom factor varies by up to +- the selected percent and is similar to crop, basically improving robustness to object size/distance variation
- Shift: translates the image horizontally/vertically
  - Shifts by up to the selected % of image dimensions. Improves robustness to imperfect centering/framing
- Shear: slants the image along an axis
  - Shears the angle within +- the selected % and is used to simulate perspective distortion

- Perspective: applies a random perspective/projective transform
  - Simulates viewing from an angle rather than straight on
- Flip: mirrors the image horizontally or vertically
  - Used to improve left-right invariance robustness